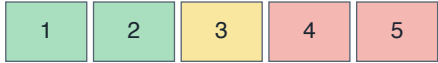


Where the platform can fail and how we would see it

Failure modes phrased for users, not engineers; high-risk rows are flagged on the left so the eye lands on what to fix first.

#	Subsystem	Failure mode	User-visible effect	Likely cause	How we would notice	S	O	D	RPN	Risk
1	Sign-in	Verification email never arrives	New users stuck in pending state	Resend throttle or spam-folder filtering	Spike in failed sign-ins; support tickets	4	2	2	16	Medium
2	Ride matching	Two riders confirmed for the last seat	One rider double-booked; manual reconciliation	Concurrent bookings without seat-level locking	Trip shows negative seats remaining	4	2	2	16	Medium
3	Cost split / payment	No in-app payment; settled by hand	No escrow protection for either side	Payment integration is design-only in v4.4	Documented gap; cited in scoping section	3	4	4	48	High
4	Safety / SOS	Distress button has no escalation path	Rider in distress lacks automated help	Backend service not yet built for SOS	Feature flag remains off; zero invocations	5	3	4	60	Critical
5	Trip-bound chat	First booking briefly fails, then succeeds	Momentary error toast, chat opens on retry	Idempotent insert path with silent retry	Backend flag for harmless duplicate	1	2	2	4	Low
6	Notification fan-out	Driver misses booking notification	Driver unaware of new rider until trip time	Partial delivery without batch transaction	Mismatch between rider and driver inboxes	3	2	2	12	Low
7	Rating system	Driver can revise rider score downward later	Retaliation risk; F5 rating-anxiety	No dispute window or rating freeze	Driver subset (N=19) flags fairness concern	4	3	2	24	Medium
8	Points / rewards	Balance always shows zero	Motivation gap; users stop engaging	Points ledger not yet provisioned in production	Points API returns empty results	3	5	1	15	Medium
9	External outreach	WeChat short-link expires before clicked	Outbound push silently fails to reach group	Link service exception is not retried	Outreach queue marked failed	3	2	2	12	Low
10	Campus whitelist	Only Cornell email addresses accepted	Multi-campus expansion blocked	Single institution domain hard-coded	Sign-up rejected for non-Cornell .edu	3	4	1	12	Low
11	Geo check-in	Spoofed GPS earns activity points	Inflated balance; fraudulent rewards	Client-trusted location with no attestation	Manual review of device metadata	2	2	3	12	Low
12	Group discovery	Community groups all start at one member	No flywheel; cold-start on community module	Discovery surface not yet exposed in app	Manual count of group membership	3	4	2	24	Medium

Severity / Occurrence / Detection



low

high

Risk tier



 left margin = high-risk row

RPN = Severity x Occurrence x Detection (each scored 1-5).